Uniswap v3 Full-Range Liquidity Provision Strategy

USDC/WETH 0.05% Pool: Trading Strategy Document

Prepared 2026-11-02 | Pending sign-off for live deployment

Executive Summary

This document specifies a passive, full-range liquidity-provision strategy on the Uniswap v3 USDC/WETH 0.05% pool. Backtested on the pool's complete real trading history, 2021-05-05 to 2026-09-08 (5.35 years), the strategy returned +10.7%, against -13.3% for a 50/50 USDC and WETH buy-and-hold benchmark and -26.7% for a 100% WETH buy-and-hold benchmark, an outperformance of +24.1 percentage points over the 50/50 benchmark. Fee income accounted for 0.251 of the 1.107 final value per dollar invested.

The strategy requires one on-chain transaction to enter and one to exit, and no active management between them. At current Ethereum network gas prices, round-trip transaction cost is approximately $1.00 under typical conditions and approximately $4.99 under elevated conditions, making the strategy cost-efficient at position sizes above approximately $499.

The backtest period includes two real market drawdowns. The strategy's maximum drawdown over the period was -49.1%, reached on 2022-06-18. A position of this kind should be sized as risk capital, with the exit criteria in Section 5 applied, not held on the assumption that the historical return repeats.

Recommendation

Proceed to a small, capital-limited live deployment (Section 4 sizing guidance; Section 5 exit criteria) only once sign-off is formally given. This document is the strategy specification for that decision, not an instruction to deploy.

1. Strategy Specification

Position: a single full-range (minimum tick to maximum tick) Uniswap v3 position in the USDC/WETH 0.05% fee-tier pool, capitalized with a 50/50 value split between USDC and WETH at entry, held without rebalancing.

Valuation (impermanent-loss exposure)

Let P_0 be the WETH/USDC price at entry and P_t the price at time t. For a full-range position, position value relative to entry value is exactly $\sqrt{P_t / P_0}$, independent of position size. This is the standard constant-product impermanent-loss identity and requires no estimation.

Fee income

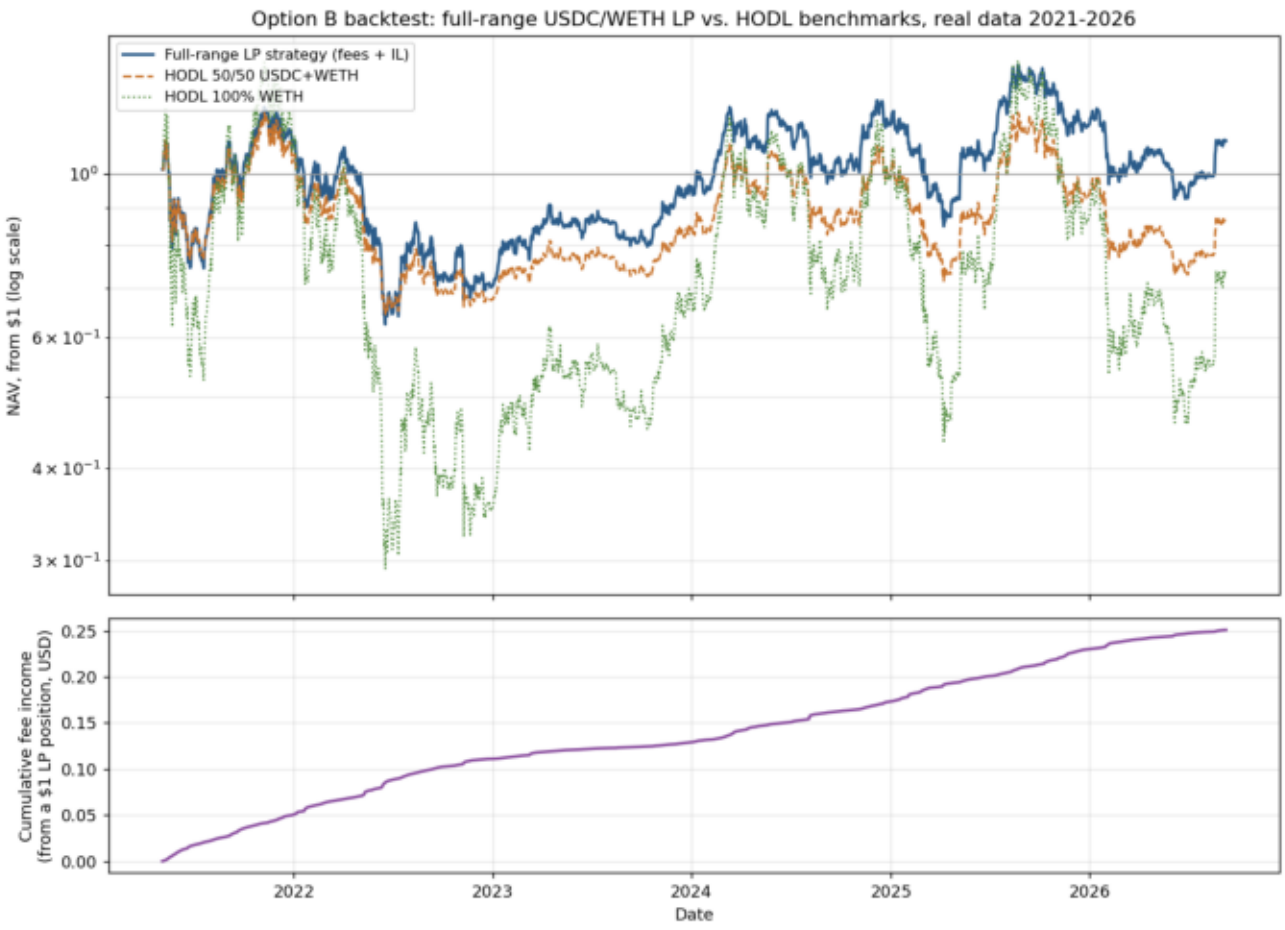
The position's liquidity contribution is fixed at entry. Its share of trading fees at any time equals its own liquidity divided by the pool's total liquidity at that time, which varies with other liquidity providers' activity. Fee income accrues automatically inside the position and compounds into its value; no separate collection transaction is required to earn it, only to realize it in a wallet.

Position sizing at entry

Entry requires swapping a portion of capital to reach a 50/50 USDC and WETH split before minting the position. This swap costs the pool's own 0.05% trading fee plus negligible price impact at the position sizes under consideration here, given the pool's typical depth (median observed total value locked $1,145,064,286 over the backtest period).

2. Backtest Result

Source: complete real swap history of the USDC/WETH 0.05% pool, 2021-05-05 to 2026-09-08 (5.35 years), 11,599,763 swaps. No synthetic or simulated data.



Strategy final value (from $1.00)	**$1.107 (+10.7%)**
50/50 HODL benchmark	**$0.867 (-13.3%)**
100% WETH HODL benchmark	**$0.733 (-26.7%)**
Cumulative fee income (from $1.00)	**$0.251**
Maximum drawdown	**-49.1% (2022-06-18)**
WETH price, start to end	**$3,402.73 → $2,494.52**

3. Transaction Cost Model

The strategy requires exactly two on-chain transactions: one to open the position, one to close it. No rebalancing transactions are required between them.

Open position (mint)	**~220,000 gas**
Close position (decrease liquidity, collect, burn)	**~180,000 gas**
Round-trip cost, typical network conditions (~1 gwei, WETH $2,495)	**$1.00**
Round-trip cost, elevated network conditions (~5 gwei, WETH $2,495)	**$4.99**

At elevated network conditions, position sizes above approximately $499 keep round-trip transaction cost under 1% of capital. Smaller positions should size this cost explicitly against expected return before entry.

4. Operational Layer

- Custody: a dedicated wallet funded only with the capital allocated to this strategy, plus a small ETH balance for gas. Hardware-wallet signing for any live deployment.
- Position management: Uniswap's NonfungiblePositionManager contract, via the Uniswap application or a scripted interface. One mint transaction at entry; one decrease-liquidity-and-collect-and-burn transaction at exit.
- Monitoring: position value and pool liquidity checked on a regular cadence (daily is sufficient for a full-range, non-rebalancing position) against a block explorer or position-tracking dashboard. No active intervention required between entry and exit.
- Accounting: fee realization and position closure are taxable events in most jurisdictions. Position-level records (entry value, exit value, fees realized) should be kept for reporting.

5. Exit Plan

Drawdown limit

Exit if position value falls more than 49% below its entry value. This is the maximum historical drawdown observed in the backtest; a live position breaching it has moved outside the strategy's tested range.

Capital-at-risk limit

Exit if the position's unrealized loss exceeds the pre-set maximum loss for this allocation, set before entry as a fraction of total trading capital, independent of the drawdown limit above.

Scheduled review

Review the position at a fixed cadence (quarterly is reasonable for a non-rebalancing position) against current pool liquidity, current fee income rate, and current gas conditions. Exit if the position's trailing fee income rate has fallen persistently below the rate needed to justify its risk, not only on a loss trigger.

Operational exit

Exit immediately on any sign of a smart-contract incident affecting the pool or the Uniswap v3 position-manager contract, independent of the position's current value.

6. Risk Disclosures

- Past performance, including this backtest, does not indicate future results. The strategy's return depends on future trading volume, fee income, and price movement in the pool, none of which are guaranteed to resemble the backtest period.
- Impermanent loss is a real, uncapped cost that moves against the position on large price moves in either direction, independent of fee income.
- The strategy carries smart-contract risk: exposure to any vulnerability in the Uniswap v3 protocol or its position-manager contract, and exposure to the underlying Ethereum network.
- Gas costs are estimated from typical and elevated 2026 network conditions and current WETH price; both are subject to change and should be re-checked before each transaction.

Sources

Backtest: complete real Uniswap v3 USDC/WETH 0.05% swap history (on-chain, 2021-05-05 to 2026-09-08). Gas unit estimates: standard Uniswap v3 NonfungiblePositionManager transaction costs. Gas price range: Ethereum mainnet daily average gas price, 2026.